Environment Energy Development

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Environment Energy Development (A09841)

Short Title: Environment Energy Development
Faculty Science and Computing
Department Science
Cluster Environmental Science
Author TWOODCOCK
Credits 5 **Level:** Advanced

Description of Module / Aims

This module examines the environmental effects resulting from the food industry. The waste management hierarchy is examined in detail. The value of food wastes is examined. The waste management strategy and carbon footprint of a food company are investigated as part of a case study. The conversion of food waste into energy is studied, especially the use of anaerobic digestion.

Programmes

`ENVI-0024`	BSc (Hons) in Food Science and Innovation (`WD_SFDSC_B`)	4 / 2 / M

Indicative Content

- Effects of food industry pollution on the environment
- Types of food waste
- Waste management hierarchy model
- Food waste as a resource, including conversion of food waste to energy
- Greenhouse gases and carbon footprint
- Legislation relating to food processing and the environment
- Energy conservation and recovery within the food sector

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Assess of the causes and effects of pollution from the food industry.
2. Establish the different food waste types, pre- and post-consumer.
3. Critique the waste management hierarchy model.
4. Evaluate the importance of greenhouse gas emissions and the carbon footprint from farm to retail food product.
5. Evaluate the conversion of food wastes to energy by various processes, in particular anaerobic digestion.
6. Appraise the legislation associated with the management of food wastes.
7. Assess energy conservation and recovery within the food sector.

Learning and Teaching Methods

- Lectures on the stated indicative content will be delivered.
- Field trips will reinforce and support theoretical elements of the course.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	2,3,4,5
Continuous Assessment	30%	
Case Studies	30%	1,6,7

Assessment Criteria

<40%: No understanding of basic principles of environmental aspects of food production.

40%-49%: Able to understand principles; may have some difficulty applying this theory.

50%-59%: Able to understand principles discussed in the course and also able to evaluate practical data and evaluate food processing from an environmental point of view.

60%-69%: In addition to the above, able to demonstrate a high level of competence and efficiency in subject area.

70%-100%: All previous to excellent level. Be able to describe and discuss in detail environmental aspects of food production and processing.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	30	
Field Trips	6	
Independent Learning	99	

Essential Material

- Arvanitoyannis, I.S. _Waste Management for the Food Industries_. London: Elsevier Academic Press, 2008.

Supplementary Material

- "Department of the Environment, Community and Local Government." http://www.environ.ie/en/
- Tiwari , B.K., T. Norton and N.M. Holden. _Sustainable Food Processing_. England: Wiley-Blackwell, 2013.
- Waldron, K. _Handbook of Waste Management and Co-Product Re__covery in Food Processing_. England: Woodhead Publishing, 2007.
- Wang , L.K., Y.T. Hung, H.H. Lo and C. Yapijakis. _Waste Treatment in the Food Processing Industry_. America: CRC Press, 2006.